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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

III Semester B. E. Regular / Supplementary Examinations Jan / Feb-2025

Common to AS / BT / CH / IM / ME

STATISTICS, LAPLACE TRANSFORM AND NUMERICAL METHODS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.
3. 2nd year Hand book of mathematics to be provided.

PART-A

			M	CO	BT
1	1.1	In a partially destroyed laboratory record of an analysis of a correlation data, the following results were observed: variance of $y = 16$, equations of lines of regression of y on x is $5y = 8x - 17$ and x on y is $5x = 2y + 14$. For the given data, value of correlation coefficient is _____ the variance of x is _____.	02	2	2
	1.2	The first three moments of a distribution about the value 2 of the variables are $1, 16$ and -40. The third moment about the mean is _____.	02	1	1
	1.3	The real and imaginary parts of the complex valued function $f(z) = e^{2z}$ respectively are _____ and _____.	02	1	1
	1.4	The coefficient of z^2 in Maclaurin's series expansion of $f(z) = \log(1 + z)$ is _____.	02	1	1
	1.5	Using Laplace transform the value of the integral $\int_0^\infty \cosh(at)\, dt, (a > 0)$, is _____.	02	2	1
	1.6	$L[t \sin(2t)]$ is _____.	02	2	2
	1.7	The frequency response of a system is found to be $\frac{1}{(9s^2-16)}$, the corresponding time response is _____.	02	1	2
	1.8	If $F(s) = \frac{1}{s^2+4s+2}$, the inverse Laplace transform of $F(s)$ is _____.	02	1	1
	1.9	The partial differential equation $\frac{\partial^2 u}{\partial x^2} - 2\frac{\partial^2 u}{\partial x \partial y} + \frac{\partial^2 u}{\partial y^2} = 0$ is classified as _____.	02	1	1
	1.10	For the one dimensional wave equation $u_{xx} = u_{tt}$ with the boundary conditions $u(0,t) = 0, u(1,t) = 0, u_t(x,0) = 0$ and the initial displacement $u(x,0) = x(1-x)$. The value of $u(0.25, 0.25)$ by taking $h = 0.25$ and $k = 0.25$ is _____.	02	2	2

PART-B

2	a	As part of a research study on software engineer salaries at a tech company have been provided with salary data (in thousands of dollars) for a set of employees. The salaries are categorized into specific ranges, with the corresponding number of employees in each range as follows:						
		Salary Range ($K) | 60 – 70 | 70 – 80 | 80 – 90 | 90 – 100 | 100 – 110 Number of Engineers | 1 | 3 | 7 | 3 | 3							
		Using this data calculate the first three central moments.	06	2	3

	b	Fit a straight line of the form $y = ax + b$ for the following data by the method of least squares. | x | 5 | 10 | 15 | 20 | 25 | | y | 16 | 19 | 23 | 26 | 30 |	05	1	2
	c	Compute the rank correlation coefficient for the following data of the marks obtained by 8 students in Engineering Mathematics (x) and Engineering Graphics (y): | x | 15 | 20 | 28 | 12 | 40 | 60 | 20 | 80 | | y | 40 | 30 | 50 | 30 | 20 | 10 | 30 | 60 |	05	2	2
3	a	Verify that $u = y + e^x \cos y$ is harmonic. If so find v such that $f(z) = u + iv$ is analytic using Milne-Thomson method.	08	2	3
	b	Obtain the Laurent's series expansion for $f(z) = \frac{z}{(z-1)(z-3)}$ in the region: i) $1 < \|z\| < 3$ ii) $\|z\| > 3$	08	2	3
		OR			
4	a	Find the Taylor's series expansion of $f(z) = \frac{1}{z^2+4}$ about $z = -i$.	08	2	3
	b	Evaluate the integral $\oint_C \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)^2(z-2)} dz$ where C is the circle $\|z\| = 3$ using Cauchy's residue theorem.	08	2	3
5	a	A periodic function $f(t)$ of period $(2\pi)/\omega, \omega > 0$ is defined by, $f(t) = \begin{cases} \sin \omega t, & 0 < t < \pi/\omega \\ 0, & \pi/\omega < t < (2\pi)/\omega \end{cases}$; Show that $L\{f(t)\} = \frac{\omega}{(s^2+\omega^2)(1-e^{-(\pi s)/\omega})}$.	08	3	3
	b	Obtain the Laplace transform of $\frac{\cos(2t) - \cos(3t)}{t} + 3(t^2+1)^2$	08	2	2
		OR			
6	a	Express the following piecewise – continuous function in terms of unit step function and hence find the Laplace transform, $f(t) = \begin{cases} 2t^2, 0 \le t < 3 \\ t + 4, 3 < t < 5 \\ 9, t \ge 5 \end{cases}$	08	3	3
	b	Find $L\{\int_0^t e^{-3t} \sin^2 2t dt + t \cos 3t \sin 2t\}$.	08	2	2
7	a	Employing convolution theorem find inverse Laplace transform of $\frac{s^2}{(s^2+9)^2}$.	08	2	3
	b	The functional form of a signal in frequency domain is given by $\frac{4s+5}{(s-1)^2(s+2)}$. Determine the function in time domain using Laplace transform.	08	2	2
		OR			
8	a	By applying Laplace transform solve the differential equation $\frac{d^2y}{dt^2} + 5\frac{dy}{dt} + 6y = 5e^{2t}$, when $y(0) = 2$ and $y'(0) = 1$.	08	2	3
	b	Find the inverse Laplace transform of the function $\frac{s+2}{s^2-2s+5} + \log\left(\frac{1-s^2}{s^2}\right)$	08	2	2

9	a	Solve $\frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}$ subject to the conditions $u(0,t) = 0, u(1,t) = 0$ and $u(x,0) = \sin(\pi x)$for $0 \le t \le 0.1$ by taking $h = 0.2, k = 0.02$. Also find the following values from the table i) u(0.2,0.04) ii) $u(0.4, 0.08)$ iii) $u(0.6, 0.06)$.	08	3	3
	b	The temperature (x, y) in a steady state heat flow in a square plate bounded by $x = 0, y = 0, x = 4, y = 4$ satisfies the Laplace equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$. Taking a square mesh of size unity and the boundary values given by $u = 0$ along $x = 0, y = 0, u = x^3$ along $y = 4, u = 16y$ along $x = 4$. Compare two iterations using Gauss Seidel iterative method.	08	3	4
		OR			
10	a	Using Schmidt's formula solve $\frac{\partial^2 u}{\partial x^2} = 2\frac{\partial u}{\partial t}$ with the conditions $u(0,t) = 0, u(12,t) = 9$ for $0 \le t \le 12$ and $u(x,0) = \frac{x}{4}(15 - x)$ for $0 \le x \le 12$ taking $h = k = 3$.	08	3	3
	b	The transverse displacement u of a point at a distance x from one end and at any time t of a vibrating string satisfies the equation $\frac{\partial^2 u}{\partial t^2} = 25\frac{\partial^2 u}{\partial x^2}$ with boundary conditions $u(0,t) = u(5,t) = 0$ and initial conditions $u(x,0) = \begin{cases} 2x, & 0 \le x \le 2.5 \\ 10 - 2x, & 2.5 \le x \le 5 \end{cases}$ and $\frac{\partial u}{\partial t} = 0$ at $t = 0$. Solve the equation numerically for half period of vibration, taking $h = 1$ and $k = \frac{1}{5}$.	08	3	4